

Experiment 5:

Aim: To demonstrate the performance of a linear regression by using chosen database with python code

Program:

```
import matplotlib.pyplot as plt
import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# Load built-in Diabetes dataset
diabetes = load_diabetes()

# Use only one feature (BMI)
X = diabetes.data[:, np.newaxis, 2]

# Target
y = diabetes.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Performance
print("R2 Score:", r2_score(y_test, y_pred))
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

# Equation
print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
print("Equation: y = {:.2f}x + {:.2f}".format(model.coef_[0], model.intercept_))

# Plot
plt.figure(figsize=(8,5))
plt.scatter(X_test, y_test, color="blue", label="Actual Data")

# Sort values so the line appears correctly
idx = np.argsort(X_test[:,0])
plt.plot(X_test[idx], y_pred[idx], color="red", linewidth=2,
        label="Regression Line")
```

```
plt.xlabel("BMI")
plt.ylabel("Disease Progression")
plt.title("Linear Regression using Diabetes Dataset")
plt.legend()
plt.show()
```

